Math 550, S20 **Name:** Answer

Quiz 8

Instructions: **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (5 points) Using the cylindrical coordinates to compute the triple integral $\iiint_W z\sqrt{x^2+y^2}\, dxdydz$ where W is the solid that lies between the cylinders $x^2+y^2=1$ and $x^2+y^2=9$, and between the planes $z=-1$ and $z=3$.

$$W^* = \{(r,\theta,z) \mid 1 \le r \le 3,\ 0 \le \theta \le 2\pi,\ -1 \le z \le 3\}$$

$$\iiint_W z\sqrt{x^2+y^2}\, dxdydz = \int_1^3 \int_0^{2\pi} \int_{-1}^{3} z \cdot r \cdot r\, dz\, d\theta\, dr$$

$$= \int_1^3 \int_0^{2\pi} r^2 \left[\frac{z^2}{2}\right]_{z=-1}^{z=3} d\theta\, dr$$

$$= \int_1^3 \int_0^{2\pi} 4r^2\, d\theta\, dr = \int_1^3 4r^2 [\theta]_{\theta=0}^{\theta=2\pi} dr$$

$$= \int_1^3 8\pi r^2\, dr$$

$$= 8\pi \left[\frac{r^3}{3}\right]_{r=1}^{r=3} = \frac{208}{3}\pi$$

Problem 2. (5 points) Find the volume of the solid $W = \{(\rho,\theta,\phi) \mid 1 \le \rho \le 2, 0 \le \theta \le \pi/2, \pi/3 \le \phi \le \pi/2\}$ using the triple integral in spherical coordinates.

$$\text{Volume} = \iiint_W \rho^2 \sin\phi\, d\rho\, d\theta\, d\phi$$

$$= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \int_1^2 \rho^2 \sin\phi\, d\rho\, d\theta\, d\phi$$

$$= \int_{\pi/3}^{\pi/2} \int_0^{\pi/2} \sin\phi \left[\frac{\rho^3}{3}\right]_{\rho=1}^{\rho=2} d\theta\, d\phi$$

$$= \int_{\pi/3}^{\pi/2} \int_0^{\pi/2} \frac{7}{3}\sin\phi\, d\theta\, d\phi = \int_{\pi/3}^{\pi/2} \frac{7}{3}\sin\phi\, [\theta]_{\theta=0}^{\theta=\pi/2} d\phi$$

$$= \int_{\pi/3}^{\pi/2} \frac{7}{6}\pi \sin\phi\, d\phi = \frac{7}{6}\pi [-\cos\phi]_{\phi=\pi/3}^{\phi=\pi/2}$$

$$= \frac{7}{6}\pi \left(0 - \left(-\frac{1}{2}\right)\right) = \frac{7}{12}\pi$$